Polynomial Continued Fractions for Logarithms and $4/\pi$: Casoratian Proofs and a Hypergeometric Non-Embedding Theorem

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Abstract

We prove two families of polynomial continued fractions (PCFs) admit exact Casoratian telescopes. The first family, with $\alpha(n) = -kn^2$ and $\beta(n) = (k+1)n + k$, converges to $1/\ln(k/(k-1))$ for all real $k > 1$, yielding geometrically convergent approximants (empirical ratio approximately $1/2$ per step for $k = 2$). The second family, with $\alpha(n) = -n(2n-3)$ and $\beta(n) = 3n + 1$, converges to $4/\pi$ and yields a rapidly convergent series for $\pi/4$ with proved error $O(2^{-N}/N^{7/2})$. We prove that the $4/\pi$ family lies outside the Gauss ${}_2F_1$ continued-fraction class: its term ratio is forced into a ${}_2F_3$ form over $\mathbb{Q}(\sqrt{5})$, placing it outside the classical hypergeometric hierarchy. Both results use a unified discrete Casoratian telescope on the underlying three-term recurrence.

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1 Introduction

Two explicit one-parameter families of polynomial continued fractions admit closed evaluations here. One family produces a full logarithmic ladder $1/\ln(k/(k-1))$ for every real $k > 1$; another yields $4/\pi$, a rapidly convergent series for $\pi/4$, and an explicit obstruction showing that the same continued fraction is not of Gauss ${}_2F_1$ type.

Classical work of Wall [12] and Lorentzen–Waadeland [8] provides the analytic background on continued fractions, convergence criteria, and the Gauss hypergeometric model relevant here. Apéry's use of a polynomial continued fraction in the irrationality proof for $\zeta(3)$ [1] shows that such recurrences can carry arithmetic content, while the Ramanujan Machine program [11, 5] has renewed the search for low-degree examples with closed values.

A *polynomial continued fraction* (PCF) with numerator polynomial $\alpha(n)$ and denominator polynomial $\beta(n)$ is

$$v = \beta(0) + \cfrac{\alpha(1)}{\beta(1) + \cfrac{\alpha(2)}{\beta(2) + \cfrac{\alpha(3)}{\beta(3) + \cdots}}}. \tag{1}$$

The convergents $C_n = p_n/q_n$ are computed via the forward recurrence

$$p_n = \beta(n)\, p_{n-1} + \alpha(n)\, p_{n-2}, \qquad q_n = \beta(n)\, q_{n-1} + \alpha(n)\, q_{n-2}, \tag{2}$$

with $p_{-1} = 1$, $p_0 = \beta(0)$, $q_{-1} = 0$, $q_0 = 1$.

The proved statements are separated carefully from the conjectural ones. Section 2 establishes the logarithmic ladder for all real $k > 1$. Sections 3 and 4 prove the base cases $m = 0$ and $m = 1$ of the π-family, and Section 6 derives the Casoratian identity for $\pi/4$. Section 7 proves non-membership in the Gauss continued-fraction class. What remains conjectural is explicit: Conjecture 3.3 proposes the general ratio law in the parameter m, and Section 5 derives the continuous Gamma quotient from that conjecture together with the proved base cases. Sections 8–10 place the proved formulas in context, and the appendices record numerical certification, exact convergents, and reproducibility data.

A companion paper [2] applies related methods to certify irrationality for an infinite parametric family of quadratic generalized continued fractions.

Related work. Euler's continued fraction for $\ln(1+x)$ [8, §2.4], Brouncker's CF for $4/\pi$ [8, §6.1], and the Ramanujan Machine conjectures [11] provide the nearest prior art. The handbooks of Lorentzen–Waadeland [8], Jones–Thron [6], and Cuyt et al. [3] are the standard references for

closed evaluations of PCFs. The (α, β) families studied here appear not to appear in those references; we note this as an observation pending a systematic comparison.

2 The Logarithmic Ladder

Theorem 2.1 (Logarithmic Ladder). *For all real $k > 1$, the PCF with $\alpha(n) = -kn^2$ and $\beta(n) = (k{+}1)n + k$ converges to $1/\ln(k/(k{-}1))$.*

Remark 2.2. The family $\alpha(n) = -kn^2$, $\beta(n) = (k{+}1)n + k$ is related to the Mercator series $\ln(k/(k{-}1)) = \sum_{j\geq 1}(k{-}1)^{-j}/j$ via the Euler series-to-continued-fraction transformation [4, §3.10(ii), eq. (3.10.5)], followed by an equivalence transformation and a reciprocal rescaling. What is new in Theorem 2.1 is the closed-form convergent expression $C_n = k/T_n$ with $T_n = \sum_{j=0}^{n}(1/k)^j/(j{+}1)$ (see equation (3) below and §4), and the unified Casoratian-style proof for all real $k > 1$.

Proof. **Claim.**

$$p_n = (n{+}1)!\,k^{n+1}, \qquad q_n = (n{+}1)!\sum_{j=0}^{n}\frac{k^{n-j}}{j+1}. \tag{3}$$

Base case. $p_0 = \beta(0) = k = 1! \cdot k^1$ and $q_0 = 1 = 1! \cdot k^0/1$.

Induction for p_n. Assume $p_{n-1} = n!\,k^n$ and $p_{n-2} = (n{-}1)!\,k^{n-1}$.

$$\begin{aligned}\beta(n)\,p_{n-1} + \alpha(n)\,p_{n-2} &= \big((k{+}1)n + k\big)n!\,k^n - kn^2(n{-}1)!\,k^{n-1}\\ &= n!\,k^n\big((k{+}1)n + k - n\big) = n!\,k^n \cdot k(n{+}1) = (n{+}1)!\,k^{n+1}.\end{aligned}$$

Induction for q_n. Write $S_n = \sum_{j=0}^{n} k^{n-j}/(j{+}1)$ and note $S_n = k\,S_{n-1} + 1/(n{+}1)$. Assume $q_{n-1} = n!\,S_{n-1}$ and $q_{n-2} = (n{-}1)!\,S_{n-2}$. Using $S_{n-1} = k\,S_{n-2} + 1/n$:

$$\begin{aligned}\beta(n)\,q_{n-1} + \alpha(n)\,q_{n-2} &= \big((k{+}1)n{+}k\big)\,n!\big(k\,S_{n-2} + \tfrac{1}{n}\big) - kn^2\,(n{-}1)!\,S_{n-2}\\ &= S_{n-2} \cdot k^2(n{+}1)\,n! + (n{-}1)!\big(kn + k + n\big).\end{aligned}$$

Meanwhile, $(n{+}1)!\,S_n = k^2(n{+}1)\,n!\,S_{n-2} + k(n{+}1)(n{-}1)! + n!$. Since $(n{-}1)!(kn{+}k{+}n) = k(n{+}1)(n{-}1)! + n!$, the two expressions are equal.

Limit. The closed forms (3) are polynomial identities in k, valid for all real k. The convergent ratio is $C_n = k/\sum_{j=0}^{n}(1/k)^j/(j{+}1)$. For $|1/k| < 1$, the series $\sum_{j=0}^{\infty} x^j/(j{+}1) = -\ln(1{-}x)/x$ converges at $x = 1/k$, giving $\sum(1/k)^j/(j{+}1) = k\ln(k/(k{-}1))$. Therefore $\lim C_n = k/(k\ln(k/(k{-}1))) = 1/\ln(k/(k{-}1))$. □

Remark 2.3. The theorem holds for all real $k > 1$, not just integers. Using Arb ball arithmetic at 120 decimal digits and forward depth $N = 2000$, we checked 15 non-integer values in $[1.5, 100]$ and found 90–120 matching digits with $1/\ln(k/(k{-}1))$. Table 1 lists representative instances.

3 The Pi Family

3.1 Base case: $2/\pi$

Theorem 3.1 (Pi Family, $m = 0$). *The PCF with $\alpha(n) = -n(2n{-}1)$ and $\beta(n) = 3n + 1$ converges to $2/\pi$.*

Table 1: Logarithmic Ladder: first instances

k	$\alpha(n)$	$\beta(n)$	Value
2	$-2n^2$	$3n+2$	$1/\ln 2$
3	$-3n^2$	$4n+3$	$1/\ln(3/2)$
4	$-4n^2$	$5n+4$	$1/\ln(4/3)$
5	$-5n^2$	$6n+5$	$1/\ln(5/4)$
φ	$-\varphi\, n^2$	$(\varphi{+}1)n{+}\varphi$	$1/\ln(\varphi/(\varphi{-}1))$
k	$-kn^2$	$(k{+}1)n{+}k$	$1/\ln(k/(k{-}1))$

Proof. **Claim.**

$$p_n = (2n{+}1)!!, \qquad q_n = (2n{+}1)!!\sum_{j=0}^{n}\frac{j!}{(2j{+}1)!!}. \tag{4}$$

Base case. $p_0 = 1 = (1)!!$ and $q_0 = 1 = 1\cdot(0!/1!!)$.

Induction for p_n. Assume $p_{n-1} = (2n{-}1)!!$ and $p_{n-2} = (2n{-}3)!!$. Using $(2n{-}1)!! = (2n{-}1)(2n{-}3)!!$:

$$\begin{aligned}\beta(n)\,p_{n-1} + \alpha(n)\,p_{n-2} &= (3n{+}1)(2n{-}1)!! - n(2n{-}1)(2n{-}3)!!\\ &= (2n{-}3)!!\big[(3n{+}1)(2n{-}1) - n(2n{-}1)\big]\\ &= (2n{-}3)!!(2n{-}1)(2n{+}1) = (2n{+}1)!!.\end{aligned}$$

Induction for q_n. Write $T_n = \sum_{j=0}^{n} j!/(2j{+}1)!!$. Note $T_{n-1} = T_{n-2} + (n{-}1)!/(2n{-}1)!!$. With inductive hypotheses $q_{n-1} = (2n{-}1)!!\,T_{n-1}$ and $q_{n-2} = (2n{-}3)!!\,T_{n-2}$:

$$\beta(n)\,q_{n-1} + \alpha(n)\,q_{n-2} = (2n{-}1)!!\big[(3n{+}1)\,T_{n-1} - n\,T_{n-2}\big].$$

Substituting $T_{n-1} = T_{n-2} + (n{-}1)!/(2n{-}1)!!$:

$$(3n{+}1)\,T_{n-1} - n\,T_{n-2} = (2n{+}1)\,T_{n-2} + \frac{(3n{+}1)(n{-}1)!}{(2n{-}1)!!}.$$

The identity reduces to $(3n{+}1)(n{-}1)! = (2n{+}1)(n{-}1)! + n!$, which holds since $n! = n(n{-}1)!$.

Limit. From the closed forms (4), we have $C_n = p_n/q_n = 1/T_n$. The identity [3, §5.2]

$$\frac{\pi}{2} = \sum_{j=0}^{\infty}\frac{j!}{(2j{+}1)!!} = \sum_{j=0}^{\infty}\frac{2^j(j!)^2}{(2j{+}1)!}, \tag{5}$$

derivable from $\arcsin(1) = \pi/2$, gives $\lim T_n = \pi/2$ and $\lim C_n = 2/\pi$. □

3.2 Parity phenomenon

Theorem 3.2 (Parity). *For even $c = 2\ell$, the PCF with $\alpha(n) = -n(2n{-}c)$ and $\beta(n) = 3n+1$ evaluates to an exact rational.*

Proof. $\alpha(\ell) = -\ell(2\ell - 2\ell) = 0$, so the CF terminates at depth $\ell{-}1$. □

The first even specialisations are listed in Table 2.

Table 2: Pi Family: even parameters (exact rationals)

c	Value	Formula
2	1	1
4	3/2	$\binom{2}{1}/2$
6	15/8	$\binom{4}{2}/2^3$
8	35/16	$\binom{6}{3}/2^4$
10	315/128	$\binom{8}{4}/2^7$
12	693/256	$\binom{10}{5}/2^8$

3.3 Odd parameters and the binomial recurrence

Conjecture 3.3 (Ratio Formula). *Define* $\mathrm{val}(m)$ *as the limit of the PCF with* $\alpha(n) = -n(2n-(2m+1))$ *and* $\beta(n) = 3n+1$. *Then* $\mathrm{val}(m+1) = \mathrm{val}(m) \cdot 2(m+1)/(2m+1)$.

Remark 3.4. Using Arb ball arithmetic at 600 decimal digits and depth $N = 5000$, the recurrence matches to at least 500 digits for $m = 0, 1, \ldots, 10$ and for the half-integer values $m = 0.5, 1.5, \ldots, 10.5$. It is consistent with the closed form $\mathrm{val}(m) = 2^{2m+1}/(\pi\binom{2m}{m})$, since $\binom{2m+2}{m+1} = \binom{2m}{m} \cdot 2(2m+1)/(m+1)$ gives ratio $4\binom{2m}{m}/\binom{2m+2}{m+1} = 2(m+1)/(2m+1)$.

Corollary 3.5 (Full Pi Family). *Assuming Conjecture 3.3, for all* $m \geq 0$ *the PCF converges to* $2^{2m+1}/(\pi\binom{2m}{m})$.

Proof. This is immediate from Theorem 3.1 and repeated application of Conjecture 3.3. Starting from $\mathrm{val}(0) = 2/\pi$, one obtains

$$\mathrm{val}(m) = \frac{2}{\pi} \prod_{j=0}^{m-1} \frac{2(j+1)}{2j+1} = \frac{2^{2m+1}}{\pi\binom{2m}{m}},$$

which is the standard central-binomial product identity. □

Table 3 records the first odd specialisations.

Table 3: Pi Family: odd parameter $c = 2m+1$

m	c	$\alpha(n)$	$\mathrm{val}(m) \cdot \pi$	$\mathrm{val}(m)$
0	1	$-n(2n-1)$	2	$2/\pi$
1	3	$-n(2n-3)$	4	$4/\pi$
2	5	$-n(2n-5)$	16/3	$16/(3\pi)$
3	7	$-n(2n-7)$	32/5	$32/(5\pi)$
4	9	$-n(2n-9)$	256/35	$256/(35\pi)$

4 Convergent Closed Forms

Theorem 4.1 ($m = 0$ convergent numerators). *For the Pi Family with* $m = 0$, $p_n = (2n+1)!! = (2n-1)!! \cdot (2n+1)$.

Proof. Base cases: $p_0=1$, $p_1=3$, $p_2=15$. Induction: assume $p_k = (2k+1)!!$ for $k < n$. The recurrence $p_n = (3n+1)p_{n-1} - n(2n-1)p_{n-2}$ with $P(n) = 2n+1$ reduces to the identity $(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-1)P(n-2)$ after dividing by $(2n-3)!!$. Expanding both sides yields $0 = 0$. □

Theorem 4.2 ($m = 1$ convergent numerators)**.** *For the Pi Family with $m = 1$ ($\alpha(n) = -n(2n-3)$, $\beta(n) = 3n+1$),*

$$p_n = (2n-1)!!\,(n^2 + 3n + 1). \tag{6}$$

The polynomial $h_n = n^2+3n+1$ (OEIS [9]) has discriminant 5 and roots $(-3 \pm \sqrt{5})/2$ related to the golden ratio.

Proof. After the substitution $p_n = (2n-1)!!\,u_n$ (using $(2n-1)!! = (2n-1)(2n-3)!!$), the recurrence reduces to

$$(2n-1)\,u_n = (3n+1)\,u_{n-1} - n\,u_{n-2}. \tag{7}$$

Substituting $u_n = n^2+3n+1$, $u_{n-1} = n^2+n-1$, $u_{n-2} = n^2-n-1$:

$$\begin{aligned}\text{RHS} &= (3n+1)(n^2+n-1) - n(n^2-n-1)\\ &= 2n^3 + 5n^2 - n - 1 = (2n-1)(n^2+3n+1) = \text{LHS}.\end{aligned}$$

□

Conjecture 4.3 (General m)**.** *For all $m \geq 0$ and fixed n, the quantity $R(n,m) = p_n(m)/(2n-1)!!$ is a polynomial in m of degree $\lfloor n/2 \rfloor$.*

A symbolic computation confirms this for $m = 0, \ldots, 12$ and $n \leq 20$. For $m \geq 2$, $R(n,m)$ is rational with denominators dividing the product of odd primes $\leq 2n-1$.

5 The Continuous Gamma Identity

The results of §§2–4 are special cases of a single continuous identity. Its extension beyond the base cases $m = 0, 1$ is conditional on Conjecture 3.3.

Theorem 5.1 (Gamma Function Representation)**.** *The formula below is proved unconditionally for $m = 0$ and $m = 1$. Assuming Conjecture 3.3, for all real $m > -\frac{1}{2}$, the PCF with $\alpha(n) = -n(2n-2m-1)$ and $\beta(n) = 3n + 1$ converges to*

$$\text{val}(m) = \frac{2\,\Gamma(m+1)}{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}.$$

Proof. We prove the identity for $m = 0$ and $m = 1$ by a purely algebraic argument, then extend to all m via the Wallis reduction formula and Conjecture 3.3 (supported numerically at 600 decimal digits and depth $N = 5000$).

Step 1: The raising operator. Define $L(f)_n = (n+2)\,f_n - (n+1)^2\,f_{n-1}$.

Lemma 5.2. *If f satisfies the m=0 recurrence $f_n = (3n+1)f_{n-1} - n(2n-1)f_{n-2}$, then $g = L(f)$ satisfies the m=1 recurrence $g_n = (3n+1)g_{n-1} - n(2n-3)g_{n-2}$.*

Proof of Lemma. Write $g_{n-1} = (n+1)f_{n-1} - n^2 f_{n-2}$ and $g_{n-2} = n\,f_{n-2} - (n-1)^2 f_{n-3}$. Expanding $(3n+1)g_{n-1} - n(2n-3)g_{n-2}$ and eliminating f_{n-3} via the m=0 recurrence at index $n-1$ yields coefficients

$$\begin{aligned}&\text{coeff. of } f_{n-1} : (3n^2+4n+1) - n(n-1) = 2n^2+5n+1,\\ &\text{coeff. of } f_{n-2} : -n^2(5n-2) + n(n-1)(3n-2) = -n(2n^2+3n-2),\end{aligned}$$

matching $g_n = (n+2)[(3n+1)f_{n-1} - n(2n-1)f_{n-2}] - (n+1)^2 f_{n-1}$ identically. □

Since L is linear, it maps the solution space $\mathcal{S}_0$ of the m=0 recurrence into $\mathcal{S}_1$. Checking initial values:

$$\begin{aligned}&L(P^{(0)})_0 = 2 \cdot 1 - 1 \cdot 1 = 1 = P_0^{(1)}, \quad L(P^{(0)})_1 = 3 \cdot 3 - 4 \cdot 1 = 5 = P_1^{(1)},\\ &L(Q^{(0)})_0 = 2 \cdot 1 - 1 \cdot 0 = 2 = 2\,Q_0^{(1)}, L(Q^{(0)})_1 = 3 \cdot 4 - 4 \cdot 1 = 8 = 2\,Q_1^{(1)}.\end{aligned}$$

By uniqueness: $P_n^{(1)} = L(P^{(0)})_n$ and $2\,Q_n^{(1)} = L(Q^{(0)})_n$.

Step 2: Poincaré growth analysis. The recurrence has Poincaré characteristic roots $\lambda = 2$ and $\lambda = 1$ (from $\lambda^2 - 3\lambda + 2 = 0$). Both $P_n^{(m)}$ and $Q_n^{(m)}$ are dominant, growing like $(2n)!!$, so $P_n^{(m)}/Q_n^{(m)} \to S^{(m)}$ at rate $O(n!/(2n)!!) = O(2^{-n}/\sqrt{n})$.

Step 3: The limit. From Step 1:

$$S_n^{(1)} = 2 \cdot \frac{(n{+}2)P_n^{(0)} - (n{+}1)^2 P_{n-1}^{(0)}}{(n{+}2)Q_n^{(0)} - (n{+}1)^2 Q_{n-1}^{(0)}}.$$

Dividing by $(n{+}2)Q_n^{(0)}$ and using $(n{+}1)^2/[(n{+}2)(2n{+}1)] \to \frac{1}{2}$, both correction terms converge to $\frac{1}{2}$, giving $S^{(1)} = 2\,S^{(0)}$.

Step 4: General m and the Gamma identity. The base case $S^{(0)} = 2/\pi$ is established by Theorem 3.1. The Wallis reduction formula gives $I_{m+1} = \frac{2m+1}{2m+2} I_m$ where $I_m = \int_0^{\pi/2} \sin^{2m}(x)\,dx = \frac{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}{2\,\Gamma(m+1)}$. Assuming Conjecture 3.3, the ratio formula $\mathrm{val}(m{+}1)/\mathrm{val}(m) = 2(m{+}1)/(2m{+}1) = I_m/I_{m+1}$ propagates inductively from $S^{(0)} = 1/I_0$, yielding $\mathrm{val}(m) = 1/I_m = 2\,\Gamma(m{+}1)/(\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2}))$ for all non-negative integers m. Under the same assumption, the identity extends to all real $m > -\frac{1}{2}$ by analytic continuation. □

Remark 5.3. For $m = 0$ and $m = 1$, Steps 1–3 give a complete algebraic proof independent of Conjecture 3.3: $S^{(0)} = 2/\pi$ and $S^{(1)} = 4/\pi$. For $m \geq 2$, the proof is conditional on the ratio conjecture. No polynomial-coefficient shift operator L_m mapping $\mathcal{S}_m$ to $\mathcal{S}_{m+1}$ has been found for general m [11].

Corollary 5.4. *Assuming Conjecture 3.3, the following specializations hold.*

(i) *Integer $m \geq 0$:* $\mathrm{val}(m) = 2^{2m+1}/\big(\pi\binom{2m}{m}\big)$.

(ii) *Half-integer $m = k{+}\frac{1}{2}$, $k \geq 0$:* $\mathrm{val}(m)$ *is an exact rational (Wallis product partial convergent).*

(iii) $\mathrm{val}(m) \sim \sqrt{4m/\pi}$ *as $m \to \infty$ by Stirling's formula.*

Proof. Part (i) is the integer case of Theorem 5.1. For part (ii), substitute $m = k{+}\frac{1}{2}$ into the Gamma quotient and use $\Gamma(k{+}1) = k!$. Part (iii) is the corresponding Stirling asymptotic for the same quotient. □

6 Casoratian Series Identity for $\pi/4$

We give a Casoratian derivation of the series identity for $\pi/4$. The telescoping argument is internal to the $m = 1$ recurrence, while the identification of the continued-fraction limit as $4/\pi$ comes independently from the Euler/Wallis analysis of the previous section. The discrete Casoratian telescope is standard machinery in the Petkovšek–Wilf–Zeilberger tradition [10]; what is novel here is the explicit companion solution constructed below.

6.1 Convergence

Proposition 6.1. *The CF* (1) *with $\alpha(n) = -n(2n{-}3)$, $\beta(n) = 3n{+}1$ converges.*

Proof. By the Śleszyński–Pringsheim theorem, the CF converges if $|\alpha(n)| \leq \beta(n)\,\beta(n{-}1)$ for $n \geq 2$. We have $\beta(n)\,\beta(n{-}1) - |\alpha(n)| = (3n{+}1)(3n{-}2) - n(2n{-}3) = 7n^2 - 2$, which is strictly positive for $n \geq 1$. □

6.2 Casoratian recursion

Let $h_n = n^2{+}3n{+}1$ (the polynomial solution from Theorem 4.2) and $g_n = q_n/(2n{-}1)!!$, both satisfying the reduced recurrence (7).

Lemma 6.2 (Casoratian recursion). *The Casoratian $W_n = h_n\, g_{n-1} - h_{n-1}\, g_n$ satisfies $W_n = \frac{n}{2n-1}\, W_{n-1}$ for $n \geq 2$.*

Proof. Substituting the recurrence (7) for h_n and g_n:

$$\begin{aligned} W_n &= \frac{(3n{+}1)h_{n-1} - n\, h_{n-2}}{2n{-}1}\, g_{n-1} - h_{n-1}\, \frac{(3n{+}1)g_{n-1} - n\, g_{n-2}}{2n{-}1} \\ &= \frac{n}{2n{-}1}(h_{n-1}\, g_{n-2} - h_{n-2}\, g_{n-1}) = \frac{n}{2n{-}1}\, W_{n-1}. \end{aligned}$$

□

Theorem 6.3 (Casoratian closed form). *For all $n \geq 1$, $W_n = n!/(2n{-}1)!!$.*

Proof. Base: $W_1 = 5 \cdot 1 - 1 \cdot 4 = 1 = 1!/1!!$. Induction: $W_n = \frac{n}{2n-1} \cdot \frac{(n-1)!}{(2n-3)!!} = \frac{n!}{(2n-1)!!}$. □

6.3 Main series identity

Theorem 6.4. *Let S be the CF with $\alpha(n) = -n(2n{-}3)$, $\beta(n) = 3n{+}1$.*

(i) *S converges, with $|S_n - S| = O(2^{-n}/n^{7/2})$.*

(ii) *$p_n = (2n{-}1)!!\,(n^2{+}3n{+}1)$.*

(iii) *$S = 4/\pi$, equivalently*

$$\frac{\pi}{4} = 1 - \sum_{k=1}^{\infty} \frac{k!}{(2k{-}1)!!\,(k^2{+}3k{+}1)(k^2{+}k{-}1)} = 1 - \sum_{k=1}^{\infty} \frac{2^k}{\binom{2k}{k}(k^2{+}3k{+}1)(k^2{+}k{-}1)}. \tag{8}$$

Proof of (iii). Part (ii) is exactly the closed numerator formula (6) from Theorem 4.2. Part (i) combines Proposition 6.1 with the tail estimate proved immediately below. It remains to establish (iii). The ratio g_n/h_n telescopes:

$$\frac{g_n}{h_n} - \frac{g_{n-1}}{h_{n-1}} = \frac{-W_n}{h_n\, h_{n-1}}.$$

Summing from $k = 1$ to n and using $g_0/h_0 = 1$:

$$\frac{g_n}{h_n} = 1 - \sum_{k=1}^{n} \frac{k!}{(2k{-}1)!!\,(k^2{+}3k{+}1)(k^2{+}k{-}1)}.$$

As $n \to \infty$, the convergents $p_n/q_n = h_n/g_n$ tend to the continued-fraction value S by Proposition 6.1. The value $S = 4/\pi$ is established independently by Lemma 5.2 and Step 3 of the proof of Theorem 5.1: the raising operator L maps $P^{(0)}, Q^{(0)}$ to scalar multiples of $P^{(1)}, Q^{(1)}$, so $S^{(1)} = 2\, S^{(0)} = 4/\pi$ by Theorem 3.1, with no appeal to the present section. Therefore $g_n/h_n \to \pi/4$, and passing to the limit in the telescoping identity yields (8). The alternative form uses $k!/(2k{-}1)!! = 2^k/\binom{2k}{k}$ (see [7, §1.2.6] for the double-factorial convention). □

Proof of convergence rate (i). By Stirling, $\binom{2k}{k} \sim 4^k/\sqrt{\pi k}$, so the general term satisfies $t_k \sim \sqrt{\pi k}/(2^k k^4) = O(2^{-k}/k^{7/2})$. Since the tail is dominated by its first term, $|S_N - \pi/4| = O(2^{-N}/N^{7/2})$. At $N = 35$ the error drops below 10^{-16}. □

7 Gauss CF Non-Membership

7.1 Factorisation over $\mathbb{Q}(\sqrt{5})$

Set $\varphi = (1+\sqrt{5})/2$, $\psi = (1-\sqrt{5})/2$. Since $(k+\varphi)(k+\psi) = k^2+k-1$ and $(k+2+\varphi)(k+2+\psi) = k^2+5k+5$, the ratio of consecutive series terms in (8) is

$$\frac{t_{k+1}}{t_k} = \frac{(k+1)(k+\varphi)(k+\psi)}{2(k+\frac{1}{2})(k+2+\varphi)(k+2+\psi)}. \tag{9}$$

7.2 Pochhammer obstruction

A standard ${}_pF_q\left(\begin{smallmatrix}\cdots\\ \cdots\end{smallmatrix}\middle| z\right)$ has term ratio equal to a product of linear factors $(k+a_i)/(k+b_j)$ with $a_i, b_j \in \mathbb{Q}$. Here the irreducible quadratics k^2+k-1 and k^2+5k+5 (both of discriminant 5) force the parameters into $\mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$.

7.3 Gauss CF non-membership

Proposition 7.1. *The CF does not arise from any ${}_2F_1$ ratio via the Gauss continued-fraction theorem.*

Proof. The Gauss CF for ${}_2F_1\left(\begin{smallmatrix}A+1, B\\ C+1\end{smallmatrix}\middle| z\right) / {}_2F_1\left(\begin{smallmatrix}A, B\\ C\end{smallmatrix}\middle| z\right)$ has unit-denominator coefficients t_k determined by $(A, B, C) \in \mathbb{Q}^3$. Solving $t_1 = c_1$, $t_2 = c_2$, $t_3 = c_3$ (where the c_k are the equivalence-transformed coefficients of our CF) yields a degree-6 polynomial system with six solutions; four are complex and two are real with $C \approx -3.259$. For these, $|t_4 - c_4| \approx 5.9$, so no match exists. □

7.4 ${}_2F_3$ identification

Remark 7.2. The term ratio (9) identifies the series as

$${}_2F_3\big(\varphi, \psi;\ \tfrac{1}{2},\ 2+\varphi,\ 2+\psi;\ \tfrac{1}{2}\big)$$

over $\mathbb{Q}(\sqrt{5})$. Since both the parameters and the function class depend on the extension $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$, the series does not reduce to any standard ${}_pF_q$ with rational parameters.

8 Euler CF Duality

The $m = 0$ CF decomposes as $S^{(0)} = \beta(0) + \alpha(1)/T = 1 - 1/T$ where T is the inner tail starting at $n = 1$. Since $S^{(0)} = 2/\pi$ (Theorem 3.1):

Proposition 8.1. $T = \pi/(\pi - 2)$, *equivalently* $T/(T-1) = \pi/2$.

Proof. $T = 1/(1 - S^{(0)}) = 1/(1 - 2/\pi) = \pi/(\pi - 2)$. The identity $T/(T-1) = \pi/2$ then follows from $T - 1 = (2)/(\pi - 2)$. □

Remark 8.2. This connects our PCF to the classical identity (5), which comes from arcsin(1) [4, §4.24.2]. The partial numerators of the $m = 0$ and $m = 1$ families differ by a linear term: $\alpha_0(n) - \alpha_1(n) = -n(2n-1) + n(2n-3) = -2n$.

9 Comparison with Brouncker's CF

Lord Brouncker's classical CF for $4/\pi$ is

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cdots}}},$$

with $\alpha^B(n) = (2n{-}1)^2$ and constant $\beta^B(n) = 2$. Both CFs converge to $4/\pi$ but differ structurally:

	Brouncker	**Novel** (m=1)
$\alpha(n)$	$(2n{-}1)^2$	$-n(2n{-}3)$
$\beta(n)$	2 (constant)	$3n{+}1$ (linear)
p_n	$(2n{+}1)!!$	$(2n{-}1)!!(n^2{+}3n{+}1)$
Series	Leibniz $\sum(-1)^k/(2k{+}1)$	Eq. (8)
Rate	$O(1/n)$	$O(2^{-n})$

	Brouncker			**Novel**		
n	p_n	q_n	S_n	p_n	q_n	S_n
0	1	1	1.000000	1	1	1.000000
1	3	2	1.500000	5	4	1.250000
2	15	13	1.153846	33	26	1.269231
3	105	76	1.381579	285	224	1.272321
4	945	789	1.197719	3045	2392	1.272993
5	10395	7734	1.344065	38745	30432	1.273166
8	34459425	28018665	1.229874	180405225	141690240	1.273237
						$4/\pi \approx 1.273\,239\,545$

At $n = 8$, Brouncker's error is $\sim 4 \times 10^{-2}$ while the novel CF error is $\sim 3 \times 10^{-6}$, a factor of 10^4. The two CFs are not related by any equivalence transformation, contraction, or Möbius substitution.

10 Convergence Theory

10.1 Worpitzky condition

Lemma 10.1. *For both families, the normalised partial numerators $|\tilde{\alpha}_n| = |\alpha_n|/(\beta_n\,\beta_{n-1})$ satisfy $|\tilde{\alpha}_n| < 1/4$ for all sufficiently large n.*

Proof. **Log family.** $|\tilde{\alpha}_n| \to k/(k{+}1)^2 < 1/4$ for $k > 1$ (since $(k{-}1)^2 > 0$).

Pi family. $|\tilde{\alpha}_n| = n(2n{-}1)/((3n{+}1)(3n{-}2)) \to 2/9 < 1/4$. At $n = 1$: $|\tilde{\alpha}_1| = 1/4$ (borderline); strict inequality holds for $n \geq 2$, and convergence at equality at finitely many n follows from the classical extension [8, §4.2]. □

10.2 Bracketing

Lemma 10.2. *If $\alpha_n < 0$ and $\beta_n > 0$ for all $n \geq 1$, and the CF converges to L, then* $\min(C_N, C_{N+1}) \leq L \leq \max(C_N, C_{N+1})$.

Proof. The determinant $p_n q_{n-1} - p_{n-1} q_n = \prod_{j=1}^{n}(-\alpha_j) > 0$ shows $C_n - C_{n-1}$ alternates in sign. See [8, Theorem 4.35]. □

11 Open Questions

The following questions arise directly from the proved theorems and remain open.

1. **Higher-degree Casoratian families.** The Logarithmic Ladder and the $4/\pi$ identity both arise from quadratic-in-n partial numerators with linear $\beta(n)$. Do analogous closed Casoratian telescopes exist for cubic $\alpha(n)$ or for higher-degree $\beta(n)$? A systematic classification of PCFs admitting a polynomial Casoratian recursion of the form $W_n = R(n)\, W_{n-1}$ with rational R would generalise both Theorem 6.3 and the Logarithmic Ladder.

2. **Generalisation of the non-${}_2F_1$ obstruction.** Proposition 7.1 excludes the $4/\pi$ family from the Gauss continued-fraction class because the term ratio factors over $\mathbb{Q}(\sqrt{5})$ rather than $\mathbb{Q}$. Does the same quadratic-extension obstruction occur for the higher members of the Pi Family ($m \geq 2$)? More generally, for which discriminant fields $\mathbb{Q}(\sqrt{D})$ do the Pochhammer factorisations (9) land, and is there a structural reason for $D = 5$ at $m = 1$?

3. **Resolving Conjecture 3.3.** The continuous Gamma identity (Theorem 5.1) is proved for $m = 0, 1$ via the linear shift operator L of Lemma 5.2. Does a polynomial-coefficient operator $L_m \colon \mathcal{S}_m \to \mathcal{S}_{m+1}$ exist for general m? An affirmative answer would discharge Conjecture 3.3 and prove the full Pi Family closed form unconditionally.

A Numerical Certification

The proved formulas were certified using Arb ball arithmetic (`python-flint` 0.8.0, 10 000-bit precision) at depth $N = 5000$. By Lemma 10.2, the limit lies in $[\min(C_N, C_{N-1}),\ \max(C_N, C_{N-1})]$. Table 4 summarises the certified enclosures for the four proved family values.

Table 4: Arb-certified intervals (depth 5000)

PCF	Target	Bracket width	Digits
$\alpha{=}{-}2n^2,\ \beta{=}3n{+}2$	$1/\ln 2$	$< 10^{-1509}$	1509
$\alpha{=}{-}3n^2,\ \beta{=}4n{+}3$	$1/\ln(3/2)$	$< 10^{-2265}$	2265
$\alpha{=}{-}n(2n{-}1),\ \beta{=}3n{+}1$	$2/\pi$	$< 10^{-1508}$	1508
$\alpha{=}{-}n(2n{-}3),\ \beta{=}3n{+}1$	$4/\pi$	$< 10^{-1518}$	1518

For the $m = 1$ CF, backward recurrence at depth 500 with 80-digit arithmetic confirms

$$S = 1.273239544735162686151070106980114896275677165923651589 98\ldots$$

matching $4/\pi$ to all 60 digits.

B Exact Convergents

Log (k=2): $p_n = (n{+}1)! \cdot 2^{n+1}$.

n	p_n	q_n	C_n
0	2	1	2.000000
1	8	5	1.600000
2	48	32	1.500000
3	384	262	1.465649
4	3840	2644	1.452345
5	46080	31832	1.447584

Limit: $1/\ln 2 = 1.442695\ldots$

Pi (m=0)**:** $p_n = (2n+1)!!$.

n	p_n	q_n	C_n
0	1	1	1.000000
1	3	4	0.750000
2	15	22	0.681818
3	105	160	0.656250
4	945	1464	0.645492
5	10395	16224	0.640688

Limit: $2/\pi = 0.636620\ldots$

Pi (m=1)**:** $p_n = (2n-1)!!\,(n^2+3n+1)$.

n	p_n	q_n
0	1	1
1	5	4
2	33	26
3	285	224
4	3 045	2 392
5	38 745	30 432
6	571 725	449 040
7	9 594 585	7 535 616
8	180 405 225	141 690 240

Limit: $4/\pi = 1.273239\ldots$

C Reproducibility

Python 3.12, `mpmath` 1.4.1, `python-flint` 0.8.0 (Arb), `sympy` 1.14.0. Forward recurrence (2) with p_{-1}=1, p_0=$\beta(0)$, q_{-1}=0, q_0=1. Runtime: <2 hours on a standard desktop.

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